SAMPLE

Earth's Sky: The Analemma

The Sun's Yearly Figure-Eight

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Introduction

Of all the shapes the sky draws for a patient watcher, the analemma is the strangest and the most personal: a figure eight, traced once per year, by the sun itself, at whatever hour the watcher chooses. It is not a constellation, not a cloud pattern, not a planet's loop against the stars, but the record of where the sun stands at the same clock time on every day of the year, stacked into a single picture. The eight appears because the sun is not where the calendar says it should be, and the two reasons it drifts, the lean of the Earth's axis and the uneven pace of its orbit, are two of the oldest problems in practical astronomy. This volume treats the analemma the way the series treats every sky subject: as a working, watchable thing, from the first recognition that the sundial and the clock disagree, through the geometry that explains the disagreement, to the craft of photographing the whole figure with nothing but a tripod, a year of patience, and a rule about clocks.

The subject earns a whole book because the analemma sits at the junction of everything else in the sky's mechanics. The figure's height is the swing of the seasons, its width is the sun's habitual unpunctuality, its lean is the watcher's own latitude and chosen hour, and its very existence is the reason sundials carry little correction tables and globe-makers print a tiny eight on the Pacific. A reader who understands the analemma understands, in one picture, obliquity and eccentricity, mean and apparent time, the equation of time, and the difference between the two ways of finding noon. Almost no other single diagram carries that much astronomy, and no other one is drawn by the sun itself, one exposure at a time, for anyone willing to keep the appointment.

The quick summary for the impatient: the analemma is the sun's yearly signature, about forty-seven degrees tall and about eight wide, skinnier at the top than at the bottom, leaning differently in every sky and every hour, reproducible with a camera and a calendar, and slowly changing its shape over the centuries as the Earth's orbit itself slowly changes. The chapters ahead show where the figure came from, what each bend means, and how to record one.

Chapter 1: The Discovery of The Analemma

The analemma has a strange discovery history, because its two halves were found separately and only met late: the sun's north-south yearly journey was charted in the deepest antiquity, the sun's east-west unpunctuality was measured with exquisite precision by the astronomers of the ancient Mediterranean and the medieval Islamic world, and the combined figure eight is really a modern object, born when clocks began to keep mean time and the sun's disagreement with them became a permanent fact of daily life.

The first half of the discovery is the seasonal one, and it is as old as astronomy itself: the sun rides higher each day to midsummer, lower each day to midwinter, and the swing was measured with shadows and horizon markers long before writing. The gnomon, the standing stick whose shadow was the sky's first instrument, made the swing visible in shadow lengths, and the oldest observatories in the archaeological record are, in effect, machines for fixing the extremes of that journey. The figure eight's vertical axis, the declination swing, is thus the most ancient measurement in the whole series.

The second half is the sun's lateness, and its discovery is one of the quiet triumphs of ancient astronomy. Careful observers noticed that the days of the year are not equal in length when measured against the stars: the time from one noon to the next varies through the year, and the seasons, counted in such days, come out unevenly. The astronomers who worked the problem out in the Hellenistic world, refining the account over generations, traced the unevenness to two causes with a correctness that still stands: the circle of the sun's yearly path is tilted against the celestial equator, so its eastward progress is sometimes diagonal, gaining or losing ground in right ascension; and the circle is not centered, so the sun's pace itself quickens and slows. A great compiler of the second century treated the geometry in a dedicated work on the analemma as a construction, a method for solving spherical problems by projection, and the name itself, from the Greek for a prop or support, attached to the figure through that tradition. This series' volume on that astronomer carries his full story; the point for this chapter is that by the ancient world's end, both ingredients of the eight were known, measured, and correctly explained, centuries before anyone drew one.

The third act is the clock. When the first good mechanical clocks began to spread, they kept, by their nature, uniform days, and uniform days do not match the sun's: the sun runs ahead of the uniform clock by a quarter hour in the autumn and behind by a similar span in the spring of the year's other half. Astronomers published the difference as tables of the equation of time, the correction to be applied to a sundial reading, and by the era of the

great national almanacs the equation was a printed column in every navigator's library. Yet even then, the equation was a table, a number for a date, and the figure eight as a drawn object was still waiting for its moment.

That moment came with mean time itself. When communities adopted mean solar time, the standard that makes every day the same length and lets clocks and timetables agree, the sun's position at any fixed clock time became a permanently wandering quantity, and the wander is the analemma. Nineteenth-century globe-makers, giving the public a picture of the new timekeeping, printed the figure on the ocean emptiness of their globes, and the little eight has decorated the Pacific of schoolroom globes ever since, most people never knowing what it is.

The last act is the camera. The first published photograph of the complete figure, a single frame carrying dozens of exposures of the sun taken across a year from one rooftop, appeared in the late 1970s and turned the diagram into a discipline and an art: a year of appointments, kept at the rate of one every week or so, at the same minute, from the same spot. The watcher's version of the discovery, and this chapter's closing point, is that the analemma is the rare sky subject that anyone can discover personally, in full, with no telescope at all, and the chapters ahead are arranged so the reader can.

A paragraph on the disagreement that started it, the two timekeepers that could not agree. The file's subject was born as a practical embarrassment: the sky's own clock, the Sun on its dial, and the human clock, the careful mechanism, disagreed, and by amounts that mattered, the shadow saying noon when the clock said ten to, the clock saying noon when the shadow was already walking away; the disagreement was not error but character, the Sun keeping its own irregular schedule, and the subject's early history is the history of tabulating the quarrel, the correction tables drawn up so the two authorities could be reconciled, the figure of the file being the quarrel itself, plotted, a full year of the Sun's refusal to run on mechanism time. The opening note worth recording: the file's origin is not an observation of the sky but a collision of standards, the moment the species owned two timekeepers good enough to catch each other cheating, and the reader should hold the origin in mind at every later chapter, the figure being less a shape in the sky than a diplomatic record, the treaty between the day and the hour, drawn as a leaning figure eight.

A second opening card, on the old word itself, a term that began as a technique. The file's name carries a history of craft: in the classical world, the term belonged to the drawing-board, the name of a geometric construction, the projection method by which instrument-makers laid out the circles of the sky on flat stone and bronze, the word meaning, roughly, the propping-up of the heavens, the taking of the spherical sky and setting it down as a figure a workman could cut; the masters of the ancient technique wrote treatises on it, and the sundials of the classical cities were its products, the word having meant the method, not the figure. The closing card for the opening: the modern meaning, the figure-eight traced by the year's noons, is a recent inheritance of an old craftsman's term, the name having wandered from the drawing board to the sky itself, and the file should savor the migration, the subject's vocabulary being one of the few in astronomy named for a workman's technique rather than a god or a discoverer, the figure overhead owing its title to the guild of dial-cutters, the sky's most abstract shape wearing the name of the most practical of arts.